

Solving Inequalities in Real-World Contexts*Additional Practice 7A.01.10*

Name _____

Date _____

Period _____

Jaxon solved the inequality $32 < -4(-1.4n - 22)$ and made more than one error. His work is shown. Use it for questions 1 through 3.

Jaxon's work

$$32 < -4(-1.4n - 22)$$

$$32 \div (-4) < (-4 \div -4)(-1.4n \div -4 - 22 \div -4)$$

$$-8 < 1(0.35n + 5.5)$$

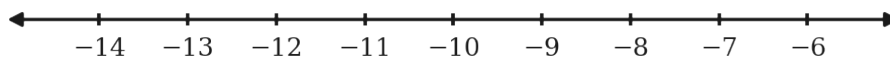
$$-8 < 0.35n + 5.5$$

$$-13.5 < 0.35n$$

$$-270/7 < n$$

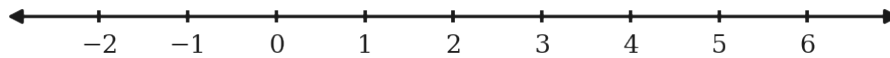
1. What errors did Jaxon make? There is more than one.
2. Write a brief note to Jaxon explaining how to avoid this in the future. Say what he did correctly as well.

3. Solve Jaxon's inequality correctly and represent the solution on the number line.

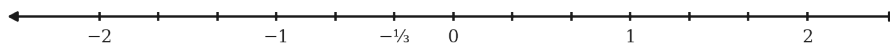


For questions 4 through 7, solve the inequality and represent the solution on the number line.

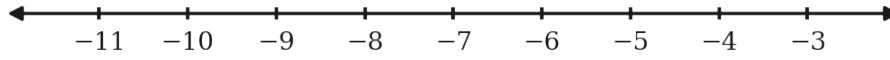
4. $(\frac{5}{3})(-a + \frac{3}{4}) \leq -2$



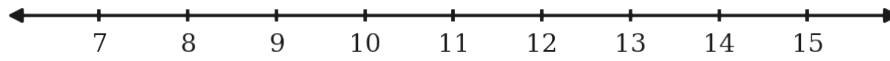
5. $-3(2t - \frac{4}{3}) > 6$



6. $(-6.2 + m)/4 > -3.3$



7. $(2/5)k - 8.4 \leq -4$



8. Explain what the $2/5$ means in this inequality.

10. Explain what the solution means in terms of the situation.